

Program : Diploma in Textile Technology	
Course Code : 2061	Course Title: Textile Fibres
Semester : 2	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2, T:1, P:0)	Periods per semester: 45

Course Objectives:

1. To impart the knowledge of natural vegetable fibres, animal and semi-synthetic fibres in terms of their structure, properties and uses.
2. To Understand the manufacture, properties & uses of nylon 66 & nylon 6

Course Prerequisites:

Topic/Description	Course code	Course name	Semester
Basic Knowledge of chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the classification and general properties of textile fibres. Illustrate the structure, properties & uses of Cotton fibre.	11	Understanding
CO2	Describe the structure, properties and uses of Bast fibres (Linen and Jute) & Animal fibres (Silk & Wool)	11	Understanding
CO3	Explain the Manufacturing process, Structure, properties and uses of Semi synthetic fibres	10	Understanding
CO4	Explain types of polymerizations and the manufacturing process, properties and uses of nylon 6.6 & nylon 6.	11	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Identify the classification and general properties of textile fibres. Illustrate the structure, properties & uses of Cotton fibre.		
M1.01	Classify textile fibres and discuss their general properties	3	Understanding
M1.02	Outline the classification, Structure, properties and uses of cotton fibres	8	Understanding
Contents: Textile Fibre: Definition and classify textile fibres on the basis of origin and chemical nature with examples. General properties of Textile fibres - Essential properties, Desirable properties. Identification of textile fibers. Cotton: Introduction - cotton producing countries, Botanical classification of cotton, Types of commercial cottons and their properties. Brief study of organic cotton & BT Cotton. Structure of cotton with sketch - longitudinal view and cross sectional view, Chemical compositions, Physical properties - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, microorganisms. Uses of Cotton.			
CO2	Describe the structure, properties& uses of Bast fibres (Linen and Jute) and animal fibres (Silk & Wool)		
M2.01	Know about the properties, extraction& uses of Linen fibers	3	Understanding
M2.02	Understand about the properties & uses of Jute fiber	2	Understanding
M2.03	Illustrate the properties & uses of silk fibre.	3	Understanding
M2.04	Discuss the properties & uses of wool fibre.	3	Understanding
	Series test - I	1	
Contents: Bast Fibres: Features of Bast fibres like Jute and Linen, Extraction of linen, Chemical composition& microscopic appearance of linen & jute, Physical and chemical properties of linen and jute.			

Silk: Introduction - varieties of silk, silk producing countries, Sericulture- Lifecycle of silk worm, Various stages involved in production of Raw Silk from cocoons, reeling with sketch, Throwing and Doubling of silk, Chemical composition of silk, Degumming and weighting of silk, Physical properties - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, micro organisms. Uses of Silk.

Wool: Introduction - wool producing countries, Commercial Varieties of wool, Classification based on Shearing of wool, Composition of wool, structure of wool with sketch, Carbonisation, Scouring and Felting of wool, Physical properties - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, micro organisms. Uses of wool. Microscopic appearance of silk and wool.

CO3	Explain the production, properties & uses of Semi synthetic fibres		
M3.01	Know the production, processing, properties & uses of Viscose rayon	4	Understanding
M3.02	Understand the production, processing, properties & uses of Acetate rayon	3	Understanding
M3.03	Know recently developed Semi-synthetic fibers	3	Understanding

Contents:

Different types of Rayons Viscose Rayon, Acetate Rayon, Cupra Rayon, Nitro Rayon - Manufacture of Viscose Rayon with flow chart, Spinning with sketch, Features of Polynosic Rayon, Physical properties of Viscose Rayon - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, micro organisms Uses of Viscose Rayon.

Various steps involved in manufacture of Acetate Rayon, preparation of dope with flow chart & spinning with sketch, Physical properties - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, micro organisms. Uses of acetate Rayon, Microscopical appearance of Viscose & Acetate Rayon.

Brief study of Lyocell, Modal, FR viscose & Bamboo fibre.

CO4	Explain the basic concepts of polymerization and manufacture of polyamide fibres		
M4.01	Introduction to manmade fibers & different types of polymerizations	3	Understanding
M4.02	Know the different methods of spinning and the drawing process.	4	Understanding
M4.03	Understand the manufacture, properties & uses of nylon 66 & nylon 6	4	Understanding
	Series Test - II	1	

Contents:

Introduction to synthetic fibres - Definition of monomers, polymers, polymerization, Degree of polymerization-Classification of Polymers based on the Structure of the Monomer Chain - Linear, branched, and cross linked polymers - Classification based on Polymerization- Condensation polymerization, Addition polymerization, Co

polymerization with their examples. Classification Based on Molecular Forces - Elastomers - Fibres - Thermoplastics - Thermosetting polymers

Principles of spinning - Melt spinning, Solution spinning - Wet spinning, Dry spinning. Importance of Drawing or Stretching- Advantages - Draw ratio - cold drawing hot drawing - Importance of Staple fiber

Man made Polyamide fiber - Nylon 6 6 - Nylon 6 - Manufacture of Nylon 6 6 from Raw materials - Polymerization Spinning - Drawing - Flow chart with equation - Microscopic appearance - Physical properties - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, micro organisms - Uses of Nylon 66.

Production of Nylon 6 from caprolactam - Condensation polymerization - Melt spinning - Stretching - Flow chart with equation – Properties - Microscopic appearance - Physical properties - length, strength, fineness, colour and chemical properties - effect of acid, alkali, water, sunlight, micro organisms - Uses of Nylon 6, Comparison of Nylon 6 with Nylon 66.

Text /Reference:

T/R	BookTitle/Author
T1	V.A Shenai Textile Fibers - .Vol.1,2
T2	S.P. Mishra- Fibre science and technology
T3	Handbook of Textile Fibres : Manmade Fibres – J Gordon Cook
R1	Gohl&Valensky -TEXTILE SCIENCSE -Mahajan publishers Ahmadabad
R2	. J T Marsh TEXTILE SCIENCE
R3	MENACHEM LEWIN - Handbook of Fiber Chemistry

Online resources:

Sl.No	Website Link
1	NPTEL :: Textile Engineering - Textile Fibres
2	https://swayam.gov.in/